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RESEARCH ARTICLE

Adaptive MPC method based on control compensation for disturbance and model parameters

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Abstract
To address the problem of performance degradation of model predictive control (MPC) for discrete-time linear systems with input–output constraints caused by external disturbances and model-plant mismatch, an adaptive MPC method based on model parameters and external disturbances control compensation is proposed. This method combines the extended state observer (ESO) with the Γ-projection algorithm to simultaneously estimate external disturbances and model mismatch parameters. A control compensation strategy is then designed using these estimates to enhance tracking performance while respecting input and output constraints. Furthermore, the boundedness of parameter estimation errors is rigorously proved based on Lyapunov stability theory. Finally, the proposed control method is applied to the liquid level control of a desorption tower in the fluidized catalytic cracking (FCC) process. The simulation results show the effectiveness and superiority of the proposed method.

Keywords MPC · External disturbance · Model-plant mismatch · Control compensation

1 Introduction

Modern industrial processes are becoming increasingly complex, imposing strict requirements on control performance, safety and economy. The traditional control methods, such as single-loop PID control, have shown limitations in dealing with multi-variable, strongly coupled and constrained dynamic systems [1]. Therefore, model predictive control (MPC), as an advanced control strategy based on model prediction, receding-horizon optimization and feedback correction, has received growing attention from both the academic community and industry due to its strong capabilities in handling multi-variable systems and constraints [2–5]. It has therefore been widely applied in various fields, including chemical processes, energy systems, and autonomous driving [6–9].

Accurate prediction of system dynamics is essential for the stable and efficient operation of MPC. However, in practical industrial applications, MPC performance is inevitably degraded by external disturbances and the model-plant mismatch (MPM), which stem from time-varying parameter uncertainties such as the attenuation of catalyst activity, environmental fluctuations and production equipment aging. To address these challenges, various adaptive and robust MPC strategies have been proposed. For example, Zheng et al. [10] proposed an adaptive MPC for Lipschitz continuous systems with nonlinear model mismatch, incorporating an event-triggered learning mechanism and proving closed-loop stability of the adaptive MPC. Wang et al. [11] developed a novel adaptive model predictive frequency control

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strategy based on the unscented Kalman filter. This adaptive MPC was realized by updating the prediction model online to capture system frequency characteristics, thereby improving disturbance rejection in power systems. Zhu et al. [12] designed an unconstrained adaptive MPC method by integrating the adaptive update law based on parameter estimation within a constrained MPC framework, which solved the MPC problem of unconstrained discrete-time linear systems with parametric uncertainties. In addition, for constrained linear systems subject to parametric uncertainties, Lu et al. [13] proposed a robust adaptive model predictive control strategy with online parameter estimation under bounded disturbances. To further mitigate the impact of external disturbances, Li et al. [14] designed an efficient robust model predictive controller for systems with bounded disturbances to enhance robust MPC by leveraging the invariant set of bounded disturbances. Sun et al. [15] employed a Kalman filter for state estimation and combined the estimates with the state gain matrix for control compensation, which effectively reduced control error fluctuations caused by various noises and disturbances under the MPC method in practice. More recently, Yang et al. [16] introduced a disturbance observer based event triggered MPC framework, where a nonlinear disturbance observer actively compensated for the disturbance, and a constant threshold type event trigger mechanism was used to correct prediction deviation caused by the remaining disturbances. This approach enhanced the robustness of standard MPC while reducing computational resource utilization for constrained nonlinear systems.

Despite these advances, the aforementioned methods typically improve the MPC from only one source of uncertainty, either external disturbances or model-plant mismatch, which limits the effectiveness of the MPC in systems subject to both simultaneously. For instance, in an MPC-based chemical production process, compensating only for feedstock disturbances while ignoring catalyst deactivation (a major cause of MPM) will lead to a progressive deterioration in temperature control accuracy. Conversely, focusing solely on MPM adaptation may result in poor rejection of external disturbances, degrading overall control performance. Note that few studies simultaneously address both parametric uncertainties and external disturbances within a unified adaptive MPC framework.

Driven by the above analysis, this paper investigates adaptive model predictive control for linear systems subject to unknown external disturbances, model uncertainty, and input/output constraints. To address these challenges, an adaptive MPC algorithm based on control compensation is proposed. The main contributions of this paper are as follows:

1. An integrated estimation scheme combining the extended state observer (ESO) and the Γ-Projection algorithm is proposed to simultaneously estimate external disturbances and uncertain parameters. The boundedness of the parameter estimation error is rigorously proved via Lyapunov stability analysis.
2. A control compensation mechanism is designed based on the estimated external disturbance and parameters, and an adaptive MPC framework incorporating this compensation is developed for systems with input and output constraints, enhancing robustness against both disturbances and model uncertainty.

2 Problem description

Consider the following uncertain linear system with external disturbances:

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) + d(k), \\ y(k) = Cx(k), \end{cases} \tag{1}$$

where $x(k) \in \mathbb{R}^l$, $u(k) \in \mathbb{R}^m$ and $y(k) \in \mathbb{R}^n$ are the state vector, control input vector and output vector, respectively. $A \in \mathbb{R}^{l\times l}$ and $B \in \mathbb{R}^{l\times m}$ are system matrices representing parametric uncertainty. $C \in \mathbb{R}^{n\times l}$ represents the constant output matrix. $d(k)$ is the smooth and bounded external disturbance.

The conventional MPC optimization problem for the nominal system subject to input and output constraints is formulated as follows:

$$\begin{aligned} &\min_{\boldsymbol{U}(k)} \sum_{i=1}^{N_p} \|y(k+i|k) - r_s(k+i)\|_Q^2 + \|\boldsymbol{U}(k)\|_R^2, \\ &\text{s.t. } \boldsymbol{x}(k+i+1|k) = A\boldsymbol{x}(k+i|k) + B\boldsymbol{u}(k+i|k), \\ &\quad y(k+i|k) = C\boldsymbol{x}(k+i|k), \\ &\quad \boldsymbol{u}_{\min}(k+i) \le \boldsymbol{u}(k+i|k) \le \boldsymbol{u}_{\max}(k+i),\ i = 0, 1, \ldots, N_c - 1, \\ &\quad y_{\min}(k+i) \le y(k+i|k) \le y_{\max}(k+i),\ i = 1, \ldots, N_p. \end{aligned} \tag{2}$$

where N_c and N_p represent the control horizon and the prediction horizon, respectively, $N_c \le N_p$; Q and R are positive definite weighting matrices for the output and control input; $r_s(k+i)$, $i = 1, 2, \ldots, N_p$ is the reference sequence; $y(k+i \mid k)$, $x(k+i \mid k)$ and $\boldsymbol{u}(k+i \mid k)$ represent the predicted output, state and control input at time $k+i$ based on information available at time k, respectively; $\boldsymbol{u}_{\min}(k+i)$ and $\boldsymbol{u}_{\max}(k+i)$ denote the lower and upper bounds on the control input at time $k+i$. $\boldsymbol{U}(k)$ denotes the predicted control input sequence over the control horizon, defined as

$$\boldsymbol{U}(k) = \begin{bmatrix} \boldsymbol{u}(k|k) \\ \boldsymbol{u}(k+1|k) \\ \vdots \\ \boldsymbol{u}(k+N_c-1|k) \end{bmatrix}_{N_c \times 1}.$$

When the system is subject to both unknown external disturbance and model–plant mismatch caused by uncertain parameters synchronously, the nominal system fails to accurately predict the system output. This inaccuracy may lead to constraint violations on outputs and control inputs of the system and reduce the effectiveness of the MPC optimization in the nominal system. As a result, the controller may fail to achieve satisfactory tracking performance. Therefore, it is necessary to design adaptive MPC with effective control compensation to ensure that the system output $y(k)$ tracks the reference signal $r_s(k)$ while satisfying all constraints.

3 Adaptive MPC design based on compensation control

To achieve desirable tracking performance while satisfying constraints on system outputs and control inputs in the presence of unknown external disturbance and model–plant mismatch, the external disturbances and system parameters are jointly estimated by combining an extended state observer (ESO) with the Γ-Projection algorithm. A control compensation term based on the estimation results is designed and incorporated into the MPC framework to obtain the optimal control law.

3.1 Disturbance estimation and parameters adaptive update

To facilitate the design of the ESO, the state vector is augmented by treating the external disturbance as an additional state. The resulting extended system is described as

$$\begin{cases} x(k+1) = \widehat{\Theta}(k)X(k) + \hat{d}(k) + L_1\big(y_c(k) - Cx(k)\big), \\ \hat{d}(k+1) = \hat{d}(k) + L_2\big(y_c(k) - Cx(k)\big), \\ y(k) = Cx(k), \end{cases} \tag{3}$$

where $\widehat{\Theta}(k) = \left[\hat{A}(k)\ \hat{B}(k)\right]$ denotes the estimated parameter matrix; $X(k) = \left[x(k)^{\mathrm{T}}\ u(k)^{\mathrm{T}}\right]^{\mathrm{T}}$ is defined as the regression vector; $L_i, i = 1, 2$ are the observer gain vectors of the ESO; $x(k)$ is the estimated system state; $\hat{d}(k)$ represents the estimate of the external disturbance; $y(k)$ is the predicted output; and $y_c(k)$ is the controlled output.

Based on the Γ-Projection algorithm [17], the parameter update law is formulated as

$$\widehat{\Theta}(k+1) = \mathrm{Proj}_{\Gamma(k)}\left(\widehat{\Theta}(k), \frac{X(k)e(k)}{c + \|X(k)\|^2}, F\right), \tag{4}$$

where $\Gamma(k)$ is a symmetric positive definite time-varying learning rate matrix; F is a convex function defining the feasible parameter set $\Omega = \{\Theta \mid |\Theta| \le \overline{\Theta}\}$, where $\overline{\Theta}$ denotes the matrix of parameter bounds; c is a known positive constant; $e(k)$ denotes the prediction error, defined as

$$e(k) = x(k) - X^{\mathrm{T}}(k-1)\widehat{\Theta}(k-1).$$

Lemma 1 *Given any $\Theta' \in \Omega$, it holds that:*

$$\left(\widehat{\Theta}(k) - \Theta'(k)\right)^{\mathrm{T}} \Gamma^{-1}(k) \left(\widehat{\Theta}(k) - \widehat{\Theta}(k-1) - \Gamma(k)\frac{X(k-1)e(k)}{c + \|X(k-1)\|^2}\right) \le 0.$$

Proof The unconstrained parameter update is as follows:

$$\Theta^*(k) = \widehat{\Theta}(k-1) + \Gamma(k)\frac{X(k-1)e(k)}{c + \|X(k-1)\|^2}. \tag{5}$$

Then project onto the convex set Ω based on Eq. (4), we have

$$\widehat{\Theta}(k) = \mathrm{Proj}_{\Gamma^{-1}(k)}\left(\Theta^*(k), \Omega\right). \tag{6}$$

For any convex set Ω and any point Θ^*, its projection $\widehat{\Theta} = \mathrm{Proj}_{\Gamma^{-1}}(\Theta^*, \Omega)$ satisfies the following variational inequality:

$$\left(\widehat{\Theta}(k) - \Theta'(k)\right)^{\mathrm{T}} \Gamma^{-1}\left(\widehat{\Theta}(k) - \Theta^*(k)\right) \le 0, \forall \Theta' \in \Omega. \tag{7}$$

Substitute $\Theta^*(k)$ into the Eq. (7), we have:

$$\left(\widehat{\Theta}(k) - \Theta'(k)\right)^{\mathrm{T}} \Gamma^{-1}(k) \left(\widehat{\Theta}(k) - \widehat{\Theta}(k-1) - \Gamma(k)\frac{X(k-1)e(k)}{c + \|X(k-1)\|^2}\right) \le 0. \tag{8}$$

Therefore, it follows that Lemma 1 holds. □

Remark 1 To ensure the estimator can track the time-varying parameters, it is necessary to assume that parameter variations are not excessively fast. Specifically, the increment of the true parameter is bounded, i.e.,

$$\|\Theta(k) - \Theta(k-1)\| \le \Delta_\Theta,$$

where Δ_Θ is a positive scalar.

Assumption 1 There exist constants $\alpha > 0$ and $T \in Z^{\dagger}$ such that for all $k \geq 0$, the following inequality holds:

$$\sum_{i=k}^{k+T} \frac{X(i)X(i)^{\mathrm{T}}}{c + \|X(i)\|^2} \geq \alpha I,$$

where I is the identity matrix.

Theorem 1 *For systems with uncertain parameters, if the regression vector satisfies persistent excitation condition stated in Assumption 1 and $\Gamma(k)$ is positive definite and bounded, then the parameter estimation error $\widetilde{\Theta}(k)$ is globally bounded.*

Proof Define the parameter estimation error as $\widetilde{\Theta} = \Theta - \widehat{\Theta}$. Based on the prediction error $e(k)$ and the parameter update law, the dynamics of the parameter error can be derived as

$$\begin{aligned}\widetilde{\Theta}(k) =& \widetilde{\Theta}(k-1) - \Gamma(k-1)\frac{X(k-1)X(k-1)^{\mathrm{T}}}{c + \|X(k-1)\|^2}\widetilde{\Theta}(k-1) \\ &+ \Gamma(k-1)\frac{X(k-1)d(k-1)}{c + \|X(k-1)\|^2} + \Delta\Theta(k) + \Delta\mathrm{Proj},\end{aligned} \tag{9}$$

where $\Delta\Theta(k) = \Theta(k-1) - \Theta(k)$, $\Delta\mathrm{Proj}$ denotes the projection correction term.

Construct the Lyapunov function as

$$V(k) = \widetilde{\Theta}^{\mathrm{T}}(k)\Gamma^{-1}\widetilde{\Theta}(k). \tag{10}$$

From Lemma 1, it follows that:

$$\begin{aligned}V(k) \leq& V(k-1) - \frac{\|X^{\mathrm{T}}(k-1)\widetilde{\Theta}(k-1)\|^2}{c + \|X(k-1)\|^2} \\ &+ \frac{2D\|X(k-)\|\|\widetilde{\Theta}(k-1)\|}{c + \|X(k-1)\|^2} \\ &+ 2\Delta_{\Theta}\|\Gamma^{-1}(k)\|\|\widetilde{\Theta}(k)\|.\end{aligned} \tag{11}$$

By recursively summing, we obtain

$$\sum_{i=k}^{k+T} V(i) \leq V(k) + \sum_{i=k}^{k+T}\left(\frac{2D^2}{c} + 2\Delta_{\Theta}\|\Gamma_{\max}^{-1}\|\|\widetilde{\Theta}(i)\|\right). \tag{12}$$

Based on Assumption 1, we obtain

$$\|\widetilde{\Theta}(k+T)\|^2 \leq \frac{1}{\lambda_{\min}(\Gamma^{-1})}\left(\frac{V(k) + T(2D^2)}{c} + 2\Delta_{\Theta}\Gamma_{\max}^{-1}D\right), \tag{13}$$

where D denotes the diameter of the feasible set, $\lambda_{\min}(\Gamma^{-1})$ is the minimum eigenvalue of Γ^{-1}. The parameter estimation error is as

$$\|\widetilde{\Theta}(k)\| \leq \sqrt{\frac{V(0) + T\left(\frac{2D^2}{c} + 2\Delta_{\Theta}\Gamma_{\max}^{-1}D\right)}{\lambda_{\min}\left(\Gamma^{-1}\right)}}. \tag{14}$$

Hence, if the regression vector satisfies persistent excitation condition stated in Assumption 1 and $\Gamma(k)$ is positive definite and bounded, then the parameter estimation error $\widetilde{\Theta}(k)$ is globally bounded. □

3.2 MPC based on control compensation

To mitigate the adverse effects of unknown external disturbances and model–plant mismatch, the control compensation equation is constructed based on the estimated external disturbance in Eq. (3), the updated parameters in Eq. (4) and parameter estimation error:

$$\widehat{B}(k)\boldsymbol{u}_c(k) + \widetilde{\Theta}(k)X(k) + \widehat{d}(k) = 0, \tag{15}$$

where $\boldsymbol{u}_c(k)$ is the control compensation term, $\widetilde{\Theta}(k)X(k)$ and $\widehat{d}(k)$ are deviations caused by model–plant mismatch and external disturbance.

Based on Eq. (15), the explicit control compensation term is obtained as follows:

$$\boldsymbol{u}_c(k) = -\left(\widehat{B}(k)\right)^{\dagger}\left(\widehat{d}(k) + \widetilde{\Theta}(k)X(k)\right), \tag{16}$$

where $\left(\widehat{B}(k)\right)^{\dagger}$ is a pseudo-inverse.

To ensure constraint satisfaction in the presence of disturbances and model uncertainty, the MPC objective function is performed subject to tightened constraints to accommodate the compensation action. The resulting objective function for the adaptive MPC scheme is given as follows:

$$\begin{aligned}\min_{\boldsymbol{U}_{\mathrm{MPC}}} & \sum_{i=1}^{N_p}\|\boldsymbol{y}(k+i|k) - r_s(k+i)\|_Q^2 + \|\boldsymbol{U}_{\mathrm{MPC}}(k)\|_R^2, \\ \text{s.t. } & \boldsymbol{x}(k+i+1|k) = \widehat{A}\boldsymbol{x}(k+i|k) + \widehat{B}\boldsymbol{u}(k+i|k), \\ & \boldsymbol{y}(k|k) = C\boldsymbol{x}(k|k), \\ & \Delta\boldsymbol{u}_{\min} \leq \boldsymbol{u}_{\mathrm{MPC}}(k+i|k) - \boldsymbol{u}_{\mathrm{MPC}}(k+i-1|k) \\ & \quad \leq \Delta\boldsymbol{u}_{\max},\ i = 1, \ldots, N_c - 1, \\ & \boldsymbol{u}_{\min} - \boldsymbol{u}_c(k+i) \leq \boldsymbol{u}(k+i|k) \\ & \quad \leq \boldsymbol{u}_{\max} - \boldsymbol{u}_c(k+i),\ i = 0, 1, \ldots, N_c - 1, \\ & \boldsymbol{y}_{\min} \leq \boldsymbol{y}(k+i|k) \leq \boldsymbol{y}_{\max},\ i = 1, \ldots, N_p,\end{aligned} \tag{17}$$

where $\Delta\boldsymbol{u}_{\min}$ and $\Delta\boldsymbol{u}_{\max}$ represent the lower and upper bounds on the control input increment, respectively.

We solve the MPC optimization problem subject to the tightened constraints given in (17) using quadratic programming [18]. The resulting optimal control sequence $\boldsymbol{U}_{\mathrm{MPC}}$ is

obtained as follows:

$$\boldsymbol{U}_{\mathrm{MPC}} = \begin{bmatrix} \boldsymbol{u}_{\mathrm{MPC}}(k|k) \\ \boldsymbol{u}_{\mathrm{MPC}}(k+1|k) \\ \vdots \\ \boldsymbol{u}_{\mathrm{MPC}}(k+N_c-1|k) \end{bmatrix}_{N_c \times 1}. \tag{18}$$

Finally, the control law implemented in the adaptive MPC scheme is constructed as the first element of the optimal control sequence and the control compensation, and is given by

$$\boldsymbol{u}(k) = \boldsymbol{u}_{\mathrm{MPC}}(k|k) + \boldsymbol{u}_c(k). \tag{19}$$

The proposed control algorithm is summarized as Algorithm 1.

Algorithm 1 Adaptive MPC based on control compensation

Input: Initial guesses $\widehat{\Theta}(0)$ and $\widehat{d}(0)$; learning rate matrix Γ; weights matrices Q and R.
Output: Optimal control law u_k at each time step k.
1: Initialize $k = 0$
2: **while** the system is running **do**
3: Update the state and estimated disturbance $\widehat{d}(k)$ using Eq. (3);
4: Update the estimated parameters $\widehat{\Theta}(k)$ using Eq. (4);
5: Calculate the control compensation $u_c(k)$ using Eq. (16);
6: Obtain the predicted control law $u_{\mathrm{MPC}}(k)$ using Eq. (17);
7: Apply optimal control law $u(k)$ from Eq. (18) to the system;
8: Let $k = k + 1$ and go to Step 2;
9: **end while**

4 Simulation experiment for desorption tower liquid level control in FCC

To verify the effectiveness of the proposed control method in this paper, the liquid level control of the desorption tower in a 2 million tons/year fluidized catalytic cracking (FCC) at the oil refinery is used as a case study. A schematic of the FCC desorption tower system is shown in Fig. 1, and the key process variables are listed in Table 1.

To evaluate the influence of adaptive algorithm parameters on control performance, a sensitivity analysis is conducted by varying the ESO bandwidth ω_0, the adaptation learning rate Γ and initial parameter $\widehat{\Theta}(0)$. We design the ESO gain vector L by solving $|sI - (A - LC)| = (s + \omega_0)^3$ using the bandwidth-based pole placement method [19]. The impact of the observer bandwidth ω_0 on the desorption tower

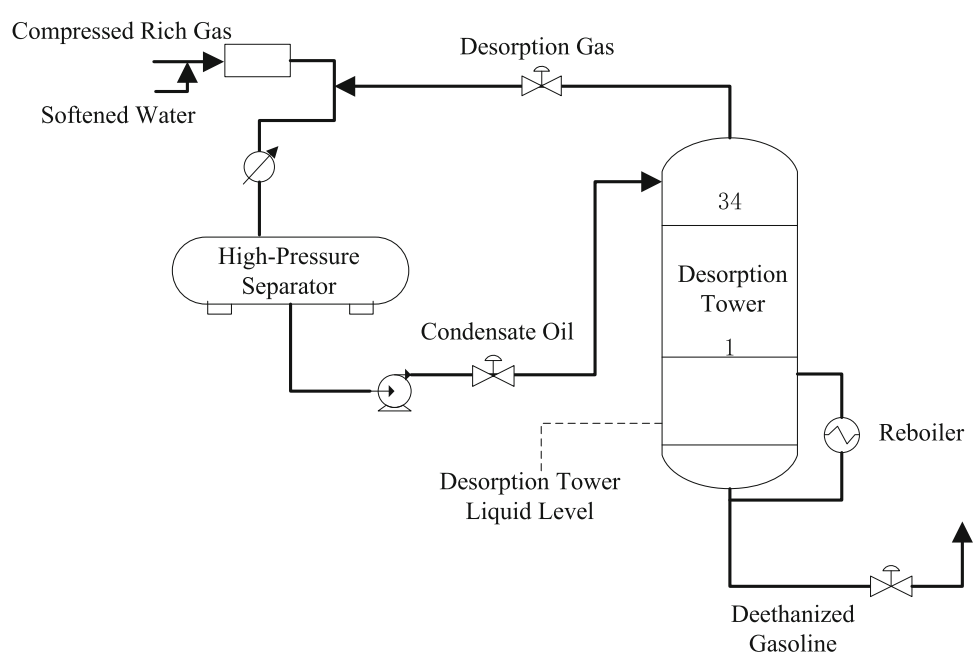

Fig. 1 Schematic of desorption tower system in FCC

Table 1 Main variables of desorption tower liquid level control system

Variables (%)	Description
u_1	The opening degree of flow valve for desorption gas
u_2	The opening degree of the three-way valve for deethanization gasoline
u_3	The opening degree of flow valve for condensate oil
y	Desorption tower liquid level

Table 2 Performance indicators of output response under different ESO bandwidths

ω_0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
t_r	8	8	8	7	7	6	5	4
$\sigma\%$	1.36	1.74	1.46	0.62	1.69	4.85	10.79	24.11
ITAE	7110.27	572.22	460.79	360.91	392.66	6469.23	5580.40	1101.31

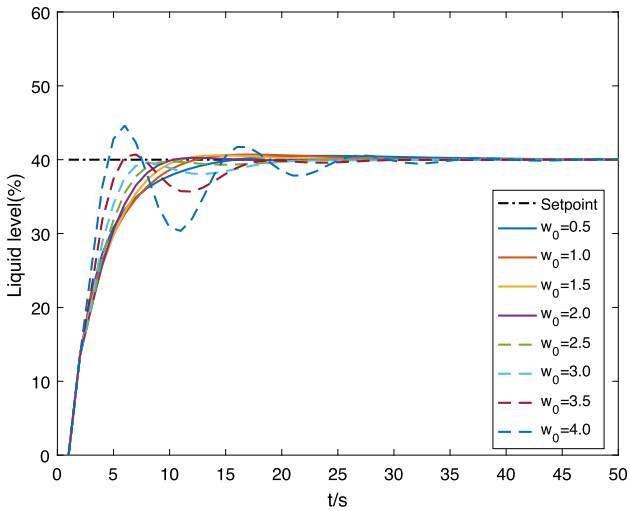

Fig. 2 Step responses under different ESO bandwidths

Table 3 Performance indicators of output response under different adaptation learning rate

Γ	0.1	1.0	3.0	5.0	10.0
t_r	17	9	7	5	4
$\sigma\%$	2.61	1.04	0.62	0.79	11.05
ITAE	839.89	331.41	360.73	380.91	926.69

liquid level control performance is evaluated in terms of overshoot $\sigma\%$, rise time t_r, and integral of time and absolute error (ITAE). Set prediction horizon $N_p = 10$, control horizon $N_c = 4$, weight matrix Q and R are diag (I_{10}), diag $(0.25 \times I_4)$. The performance metrics for the ESO bandwidth ω_0 are summarized in Table 2, and the corresponding step responses are shown in Fig. 2.

As shown in Fig. 2 and Table 2, increasing the observer bandwidth ω_0 results in a faster system response. However, excessively high bandwidth degrades stability and control accuracy due to noise amplification and overshoot. We determine that the optimal ESO bandwidth as $\omega_0 = 2.0$ based on a comprehensive evaluation of performance indices. Similarly, the adaptation learning rate Γ is tuned between 0.1 and 10.0; the influence of Γ on control performance is summarized in Table 3.

As shown in Table 3, if the Γ is too small, the parameter update process becomes sluggish, leading to slow compensation for model-plant mismatch and disturbances. This results in poor transient performance, such as large tracking errors or slow response. Conversely, an excessively large Γ causes overly aggressive parameter adaptation, which may excite high-frequency dynamics, resulting in oscillatory control signals and large overshoot. Therefore, a moderate Γ is essential to balance convergence speed and robustness. Based on simulation results, the Γ is recommended to be selected within the range [1, 5] as a practical tuning guideline for achieving satisfactory performance.

Table 4 Performance indicators of transient response under different $\widehat{\Theta}(0)$

$\widehat{\Theta}(0)$	–30%	–15%	15%	30%
t_r	7	5	4	8
$\sigma\%$	2.25	1.86	0.35	1.79
ITAE	639.89	601.41	583.61	889.01

Table 4 summarizes key performance indicators for different initial parameters $\widehat{\Theta}(0)$ with deviations of $\pm 15\%$ and $\pm 30\%$ from the nominal model parameters. The results show that while the algorithm remains stable across a wide range of initial values, the transient response (particularly overshoot and convergence speed) is sensitive to the choice of $\widehat{\Theta}(0)$. In practical applications, we recommend setting $\widehat{\Theta}(0)$ based on prior knowledge or pre-experiments. Specifically, open-loop step tests are conducted to estimate the static gain and dominant time constant, which are then used to initialize the parameter vector. When prior data is unavailable, a conservative choice is to set $\widehat{\Theta}(0)$ to mid-range of expected parameter variation. This approach ensures faster convergence and mitigates excessive transient oscillations.

To verify the effectiveness and superiority of the proposed MPC (ESO-AMPC) method, the liquid level set-point tracking performance is tested under various model-plant mismatches, including 5%, 10%, -15% and -20% model deviations from the nominal model. Additionally, external disturbances are introduced at the 80th and 130th sampling instants to assess disturbance rejection capability. The proposed ESO-AMPC is compared with the MPC method from [20] under identical operating conditions. The simulation results are shown in Fig. 3.

As shown in Fig. 3, under different degrees of model–plant mismatch, the conventional MPC achieves steady-

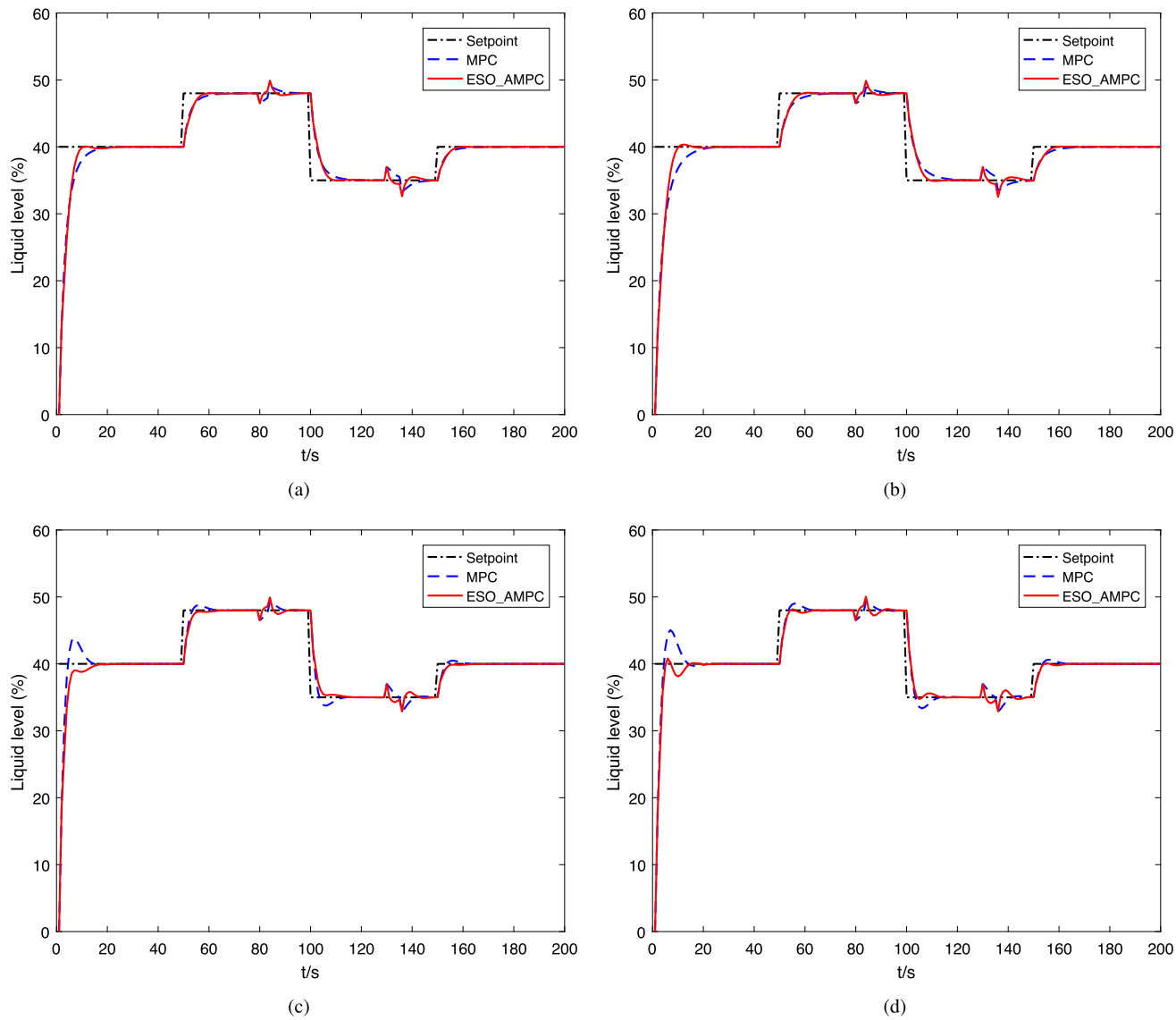

Fig. 3 Setpoint tracking control results of liquid level with different control methods. **a** 5% model–plant mismatch. **b** 10% model–plant mismatch. **c** −15% model–plant mismatch. **d** −20% model-plant mismatch

state tracking of the reference liquid level and demonstrates good robustness against external disturbances, but the system exhibits significant overshoot under −15% and −20% model-plant mismatch, slow response under 5 and 10% model-plant mismatch. In contrast, the proposed ESO-AMPC with control compensation stabilizes the liquid level quickly without overshoot. Compared with MPC, the dynamic performance of the system is significantly improved. Furthermore, under external disturbances at $t = 80\,\mathrm{s}$ and $t = 130\,\mathrm{s}$, the proposed method is superior to the MPC in both response speed and control accuracy. In conclusion, under simultaneous model-plant mismatch and external disturbances, the proposed adaptive MPC method based control compensation enhances dynamic performance of the system, demonstrating superior robustness and effectiveness.

5 Conclusion

In this paper, an adaptive MPC strategy is developed for linear systems subject to input–output constraints under external disturbances and model–plant mismatch. By integrating an extended state observer (ESO) for disturbance estimation and a Γ-projection-based parameter adaptation mechanism, the proposed method effectively compensates for uncertainties and enhances control performance. Theoretical analysis confirms the boundedness of prediction parameter errors, ensuring algorithmic stability under the

assumption of uncertain systems. The effectiveness of the approach is demonstrated through simulation studies on the liquid level control of a desorption tower in a fluid catalytic cracking (FCC) process. The proposed controller exhibits superior disturbance rejection and constraint handling compared to conventional MPC.

This study assumes that only the system matrices (A, B) of the prediction model are uncertain, while the output matrix (C) is considered known. Simultaneous estimation of (A, B, C) would require stronger persistence of excitation conditions, which are beyond the current scope. Future work will focus on extending the adaptive estimation framework to incorporate adaptation of the output matrix C, thereby further enhancing the robustness of the control system.

Data Availability The data analyzed during this study are included in this published article. However, the underlying modeling data that support the findings of this study are not publicly available due to commercial confidentiality restrictions.

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